Instability #21: $n = 2^{d_s}$ Refuted by Simulation

$n = 4$ Gives $d_s \approx 3.6$, Not 2; $|\beta|/d_s$ Is Not Universal; and the Honest Final Boundary

The $n = 2^{d_s}$ path is closed. The bracket identity bracket $= \kappa/\ln 2$ is the remaining statement.

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Abstract

V20.3 was run at $n = 4$ (2 seeds, 120 sweeps each) to test the "Saturation Identity" $n = 2^{d_s}$. Result: $d_s \approx 3.58$ (not 2.0 as required), $|\beta| \approx 1.85$. The theorem is **refuted**.

Two findings from this test: (1) d_s is anchored near 3 by the 3D X-space geometry regardless of n; changing n from 8 to 4 shifts d_s from 3.03 to 3.58, not to 2. (2) $|\beta|/d_s = 0.516$ at $n = 4$ vs $1/(2\ln 2) = 0.721$ at $n = 8$; the ratio is *not* universal across n.

This refutes both the Saturation Identity and the stronger claim that $|\beta| = d_s/(2\ln 2)$ is independent of n. The formula holds at $n = 8$ (where the bracket identity is confirmed at 0.23%), but the bracket identity itself depends on n through $H_{\text{after}} = \ln n - D_{\text{KL}}$.

Instability #21 honest status: **specific to $n = 8$, $d_s = 3$**. The bracket identity bracket $= \kappa/\ln 2$ was derived and confirmed for V20.3 as configured; its range of validity (other n, other d_s) is unknown.

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1 The $n = 4$ Test

V20.3 at $n = 4$: Results (2 seeds, 120 sweeps each)

Config	d_s	$\lvert\beta\rvert$	$\lvert\beta\rvert/d_s$	$1/(2\ln 2)$	gap
$n = 8$	3.03 ± 0.08	2.212	0.730	0.721	1.2%
$n = 4$	3.58 ± 0.10	1.847	0.516	0.721	28.5%

Prediction from Saturation Identity: $n = 4 \to d_s = 2$, $|\beta| = 1.443$.
Actual: $d_s = 3.58$, $|\beta| = 1.85$.

Confirmed: $n = 2^{d_s}$ is Refuted

- d_s at $n = 4$ is 3.58, not 2.0 (+79% from prediction).
- d_s is anchored by X-space (3D geometry); changing n does not move d_s to $\log_2 n$.
- $|\beta|/d_s$ is not universal: 0.730 at $n = 8$, 0.516 at $n = 4$.

The Saturation Identity is empirically refuted. CRF AI's claim of "Theorem Level" closure was premature.

2 Why $|\beta| = d_s/(2\ln 2)$ Holds at $n = 8$ Only

The formula $|\beta| = d_s/(2\ln 2)$ requires both:

$$f_{12} = \frac{d_s}{2} \quad \text{and} \quad \langle \text{bracket} \rangle_{\text{ev}} = \frac{\kappa}{\ln 2}$$

At $n = 8$: $f_{12} = 3/2$ (Kuramoto sub-threshold, universal for any n, but equals $d_s/2 = 3/2$ only when $d_s = 3$).

Why the $n = 4$ Result Breaks the Formula

At $n = 4$: $d_s \approx 3.58$, so $d_s/2 \approx 1.79$. But $f_{12} = 3/2 = 1.5$ (still from Kuramoto, $\phi_{\text{var}} = 1/2$). The gap $f_{12} \neq d_s/2$ immediately means $|\beta| \neq d_s/(2\ln 2)$.
At $n = 8$: $d_s = 3.03 \approx 3$, so $d_s/2 \approx 1.52 \approx f_{12} = 1.527$. The near-equality $f_{12} \approx d_s/2$ is what makes the formula hold.
Root cause: the Kuramoto coupling gives $f_{12} = 1.5$ regardless of n. The spectral dimension gives $d_s/2 \approx 1.5$ at $n = 8$ (because $d_s \approx 3$). At $n = 4$, $d_s \approx 3.6$, so $d_s/2 \approx 1.8 \neq 1.5$. The two drift apart.

The bracket at $n = 4$ also changes because $H_{\text{after}} = \ln n - D_{\text{KL}} = \ln 4 - D_{\text{KL}}$ is lower than at $n = 8$, shifting bracket away from $\kappa/\ln 2$.

3 What Is Specific to $n = 8$

The Coincidence at $n = 8$, $d_s = 3$

At $n = 8$, $d_s = 3$:

- $f_{12} = 3/2$ from Kuramoto (universal).
- $d_s/2 = 3/2$ from X-space geometry (specific to 3D).
- $f_{12} = d_s/2$: true at $n = 8$ because both equal $3/2$.
- bracket $= \kappa/\ln 2$: confirmed at 0.23% for this specific configuration.

The formula $|\beta| = d_s/(2\ln 2)$ holds at $n = 8$ as a result of $f_{12} = d_s/2$ (accidental equality at $d_s = 3$) and bracket $= \kappa/\ln 2$ (confirmed numerically).

Whether it holds for other n and d_s requires further investigation. At $n = 4$ ($d_s \approx 3.58$): it does not hold.

4 Revised Honest Boundary of Instability #21

Result	**Status**	**Note**
bracket $= \kappa/\ln 2$ at $n = 8$	Confirmed	0.23%; specific to $n = 8$
$f_{12} = 3/2$	Derived	Kuramoto; any n
$f_{12} = d_s/2$ at $n = 8$	Confirmed	Because $d_s \approx 3$
$\lvert\beta\rvert = d_s/(2\ln 2)$ at $n = 8$	Confirmed	3.4% finite-N
$n = 2^{d_s}$ as theorem	Refuted	$n = 4$ gives $d_s = 3.6 \neq 2$
$\lvert\beta\rvert/d_s = 1/(2\ln 2)$ universal	Refuted	Fails at $n = 4$ by 28%
bracket $= \kappa/\ln 2$ general	Unknown	Not tested at $n \neq 8$
Analytic proof of bracket identity	Open	For $n = 8$, $d_s = 3$ specifically

Honest Summary

For V20.3 as configured ($n = 8$, 3D X-space, $d_s \approx 3$):

$$|\beta| = \frac{d_s}{2\ln 2} = \frac{3}{2\ln 2} \approx 2.164$$

is confirmed numerically to 3.4%.
This formula is *specific to the V20.3 configuration*. It does not hold universally across different n.
The remaining open question for Instability #21: prove analytically that bracket $= \kappa/\ln 2$ for the specific case $n = 8$, $d_s = 3$, from the δ-chain. This does not require $n = 2^{d_s}$; it requires a derivation of the M-averaging identity for V20.3's actual dynamics.

The formula held at $n = 8$. It did not hold at $n = 4$. This is not a failure — it is information. It tells us that the Law of Exchange is not written in the abstract; it is written in the specific architecture of V20.3. Whether that architecture is the only one consistent with $d_s = 3$ is a question the cascade has not yet answered.